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4th Feb 2026 | DA6401

- $\frac{\partial L}{\partial w_{jk}^{l}}$, $\frac{\partial L}{\partial b_{k}^{l}}$ ⇒ recap of back propagaⁿ

- Vectorizaⁿ → Why we do the outer product?

~~→ Automatic differentiaⁿ~~

- In a fully connected layer, we don't update the wts 1 by 1. We try to update the entire wt. matrix W^l at once.

Individual ⇒ $\frac{\partial L}{\partial w_{ij}} = \delta_j \cdot x_i$

Matrix ⇒ $\boxed{\nabla_w L = \delta x^T}$

- Since, $z_i = \sum_j w_{ij} a_j$, the partial derivative $\frac{\partial z_i}{\partial w_{ij}}$ is only non-0 (a_j) when the indices match.
- This maps every combinaⁿ of neuron i & input j into its corresponding slot in the wt. matrix.

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- General Backpropagaⁿ

1. Output layer (layer L)

$$\boxed{\delta^L = \hat{y} - y}$$

2. Hidden layers ($l = L-1, \dots 2$)

$$\boxed{\delta^l = (W^{l+1})^T \delta^{l+1} \odot \sigma'(z^l)}$$

element-wise multiplicaⁿ

3. Wt. updates (for all layers)

$$\boxed{\frac{\partial L}{\partial W^l} = \delta^l (a^{l-1})^T}$$